

UNIT - 5	9 Hours
Overview of Electrical Machines, Protective Devices	
Magnetic Circuits Concepts, dot convention, Transformers -Working Principle, Types, Applications	
Electrical Machines: DC Motor, Stepper Motor - working principle and applications	
Protective devices-Fuse, MCB	
Earthing – Necessity, Types, Ground and neutral concept	

5.1 Magnetic Circuits Concepts

Introduction to magnetic circuits:

A magnetic circuit is a closed path followed by magnetic flux, analogous to an electric circuit's path for electric current. It consists of magnetic materials of high permeability like iron or steel, often with air gaps, and is used to direct and control magnetic fields. These circuits are found in devices like electric motors, transformers, and relays.

Consider the magnetic circuit shown in Fig.5.1

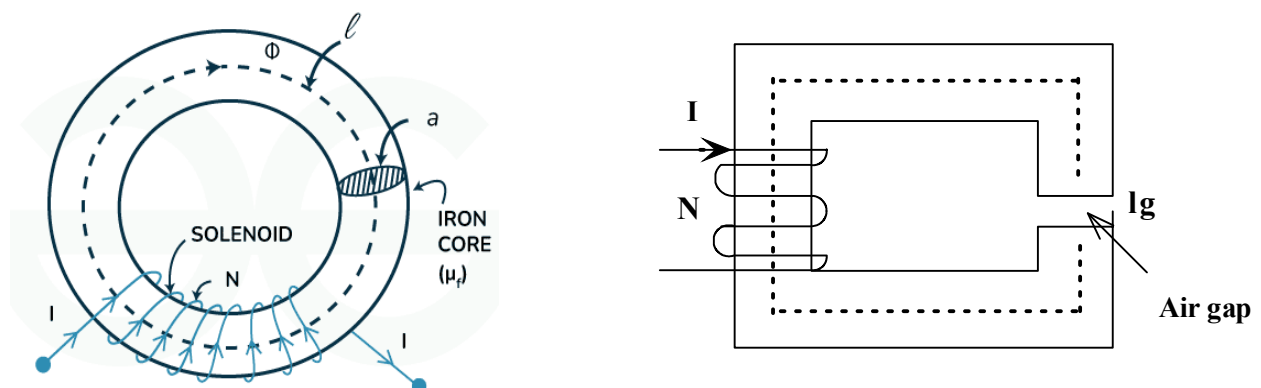


Fig.5.1. Magnetic circuits

N = number of turns

a = area of cross section in m^2

ℓ = circumferential length in m

Φ = magnetic flux in Wb

I = current in A

[Type here]

The operation of a magnetic circuit relies on the principle that a current flowing through a coil creates a Magnetomotive force (MMF), which drives magnetic flux through the circuit. The opposition to this flux is called reluctance, analogous to resistance in an electrical circuit.

Terms used in magnetic circuit:

Magnetic Core:

The primary material within the magnetic circuit, often made of iron or steel that provides a path for the magnetic flux. Examples: Air, Iron, Ferrite, Copper

Air Gap:

A non-magnetic space within the magnetic circuit, often introduced to control the magnetic field or create a specific magnetic effect.

Magnetic Field:

A magnetic field is a region around a magnet. Magnetic field lines are used to visualize the field.

Magnetic field strength:

It is often denoted by H , is a measure of the magnetizing force that produces a magnetic field in a material. It's a vector quantity, meaning it has both magnitude and direction. The SI unit for magnetic field strength is the ampere per meter (A/m).

Magnetomotive Force (MMF):

The driving force that creates magnetic flux, analogous to electromotive force (EMF) in an electrical circuit. It is generated by a current (I) flowing through a coil with a certain number of turns (N), expressed as $MMF = I * N$.

The unit of MMF is ampere-turns (AT).

Magnetic Flux:

The magnetic field lines existing in a magnetic field is the magnetic flux Φ . It is analogous to current in an electrical circuit. It's directly proportional to the MMF and inversely proportional to the reluctance. Its unit is weber and symbol is Wb.

Magnetic flux density:

It is defined as the amount of magnetic flux passing through a unit area perpendicular to the magnetic field lines (B). It is also known as magnetic induction, is a measure of the

strength and direction of a magnetic field. It is defined as the magnetic flux per unit area.

$$\left(B = \frac{\phi}{a} \right). \text{ Its unit is Wb/m}^2.$$

Reluctance:

The opposition to the establishment of magnetic flux in a magnetic circuit is called Reluctance of the circuit $\mathcal{R}$. It is analogous to resistance in an electrical circuit. Reluctance depends on the material's permeability (μ), the length (ℓ) of the magnetic path, and the cross-sectional area (a) of the path. Its unit is Ampere-turns per Weber (AT/Wb).

Reluctance of the magnetic circuit can be calculated using the equation $\mathcal{R} = \frac{\ell}{\mu a}$

Permeability:

Permeability (μ) refers to the ability of a material to allow magnetic field to pass through it. It is similar to conductivity in electrical circuits. It is given by $\mu = \mu_0 \mu_r$

μ_0 = absolute permeability ($4\pi \times 10^{-7}$ H/m (Henry per meter)

μ_r = relative permeability (free space (vacuum)=1, Copper=0.999, wood=1, Iron=5500)

It is also defined as the ratio of Magnetic Flux Density to Magnetic Field Strength.

$$\mu = \frac{B}{H}$$

Analogy between Magnetic and Electric Circuits:

Sl.No	Magnetic Circuit	Electric Circuit
1	MMF in AT	EMF in Volts
2	Flux exists (Wb)	Current flows (A)
3	Flux Density ($B = \frac{\phi}{a}$)	Current Density ($J = \frac{I}{a}$)
4	Reluctance ($\mathcal{R} = \frac{\ell}{\mu a}$)	Resistance ($R = \zeta \frac{\ell}{a}$)
5	Permeance ($P = \frac{1}{\mathcal{R}}$)	Conductance ($G = \frac{1}{R}$)

Applications:

Magnetic circuits are fundamental to the operation of various electrical and electromechanical devices.

Examples include:

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Electric motors: Magnetic circuits create the magnetic fields that drive the rotation of the motor's rotor. Ex: DC Motor, Induction Motor

Transformers: Magnetic circuits transfer electrical energy between circuits through magnetic coupling.

Relays: Magnetic circuits use magnetic force to control the switching of electrical circuits.

Generators: Magnetic circuits are used to induce voltage by moving conductors within a magnetic field. Ex: DC generator, Synchronous Generator

Galvanometers: Magnetic circuits are used to measure electric current.

In essence, magnetic circuits are the pathways through which magnetic fields travel, and understanding them is crucial for designing and analysing a wide range of electromagnetic devices

Analysis of Series Magnetic Circuits:

In a series magnetic circuit, the same flux ϕ flows through each part of the circuit.

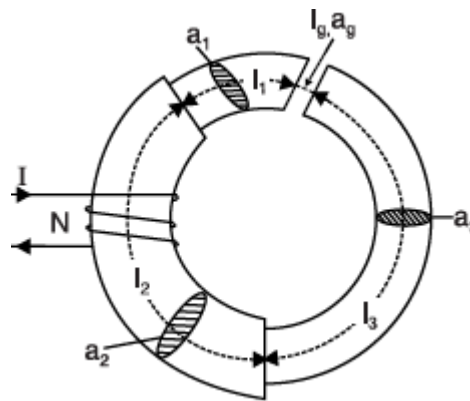


Fig.5.2. Series magnetic circuit

Consider a ***composite series magnetic circuit** consisting of three different magnetic materials of different relative permeability along with an air gap as shown in Fig.5.2. Each part of this series magnetic circuit will offer reluctance to the magnetic flux Φ . The reluctance offered by each part will depend upon dimensions and μ_r of that part. Since the different parts of the circuit are in series, the total reluctance is equal to the sum of reluctances of individual parts, *i.e.*

Step-1: To determine reluctance and MMF of each material

$$\text{Reluctance of material-1: } \mathcal{R}_1 = \frac{l_1}{a_1 \mu_0 \mu_{r1}}$$

$$\text{Magnetomotive Force (MMF1) of material -1: } \text{MMF}_1 = \Phi \times \mathcal{R}_1 = \Phi \left[\frac{l_1}{a_1 \mu_0 \mu_{r1}} \right]$$

[Type here]

(5.1)

$$\text{Reluctance of material-2: } \mathcal{R}_2 = \frac{l_2}{a_2 \mu_0 \mu_{r2}}$$

$$\text{Magnetomotive Force (MMF}_2\text{) of material-2: } \text{MMF}_2 = \oint \times \mathcal{R}_2 = \oint \left[\frac{\ell_2}{a_2 \mu_0 \mu_{r2}} \right]$$

(5.2)

$$\text{Reluctance of material-3: } \mathcal{R}_3 = \frac{\ell_3}{a_3 \mu_0 \mu_{r3}}$$

$$\text{Magnetomotive Force (MMF}_3\text{) of material-3: } \text{MMF}_3 = \oint \times \mathcal{R}_3 = \oint \left[\frac{\ell_3}{a_3 \mu_0 \mu_{r3}} \right]$$

(5.3)

$$\text{Reluctance of air gap: } \mathcal{R}_g = \frac{\ell_g}{a_g \mu_0 \mu_{rg}} = \frac{\ell_g}{a_g \mu_0 (1)} = \frac{\ell_g}{a_g \mu_0} \quad (\mu_{rg} = 1)$$

$$\text{Magnetomotive Force (MMF-g) of air gap: } \text{MMF}_g = \oint \times \mathcal{R}_g = \oint \left[\frac{\ell_g}{a_g \mu_0} \right]$$

(5.4)

Step-2: To determine total MMF

$$\text{Total MMF} = \text{MMF}_1 + \text{MMF}_2 + \text{MMF}_3 + \text{MMF}_g \quad (5.5)$$

Substituting equation (5.1) to (5.4) in (5.5)

$$\text{Total MMF} = \oint \left[\frac{\ell_1}{a_1 \mu_0 \mu_{r1}} \right] + \oint \left[\frac{\ell_2}{a_2 \mu_0 \mu_{r2}} \right] + \oint \left[\frac{\ell_3}{a_3 \mu_0 \mu_{r3}} \right] + \oint \left[\frac{\ell_g}{a_g \mu_0} \right]$$

(5.6)

Exercise 1:

An iron ring of cross-sectional area 6 cm² is wound with a wire of 100 turns and has a saw cut of 2 mm as in Fig.5.3 Calculate the magnetizing current required to produce a flux of 0.1 mWb if mean length of magnetic path is 30 cm and relative permeability of iron is 470.

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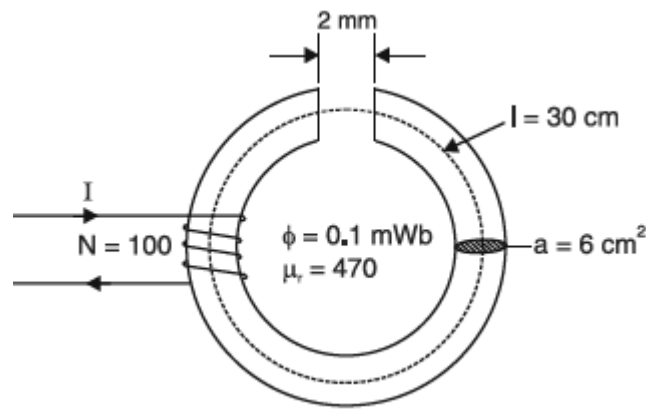


Fig.5.3: Exercise 1

Solution:**Step-1:** Note down given dataNumber of turns= $N=100$ Length of iron= $l_i=30 \times 10^{-2}$ mFlux= $\Phi=0.1$ mWbArea of cross section of iron= $a_i=6 \times 10^{-4}$ m²Relative Permeability for the air gap= $\mu_0=4\pi \times 10^{-7}$ H/mRelative Permeability for iron= $\mu_r=470$ Length of air gap= $l_g=2 \times 10^{-3}$ mArea of cross section of air gap= $a_i=6 \times 10^{-4}$ m²Relative Permeability for air gap= $\mu_0=1$ **Step-2:** Calculate MMF require for the iron and airgap

M M F r e q u i r e d f o r i r o n m a t e r i a l =

$$\Phi \left[\frac{\ell_i}{a_i \mu_0 \mu_{ri}} \right] = 0.1 (10^{-3}) \left[\frac{30 \times 10^{-2}}{6 \times 10^{-4} (4\pi \times 10^{-7}) \times 470} \right] = 84.83 \text{ AT}$$

M M F r e q u i r e f o r a i r g a p =

$$\Phi \left[\frac{\ell_g}{a_g \mu_0 \mu_{rg}} \right] = 0.1 (10^{-3}) \left[\frac{2 \times 10^{-3}}{6 \times 10^{-4} (4\pi \times 10^{-7}) \times 1} \right] = 3.51 \text{ AT}$$

Step-3: Calculate total MMFTotal MMF=MMF required for iron material+ MMF require for air gap= $84.83+3.51=350.63$ AT**Step-4:** Determine magnetizing current required $NI=350.63$ $100I=350.63$

[Type here]

$$I = 350.63/100$$

$$I = 3.50 \text{ A}$$

Analysis of Parallel Magnetic Circuits:

A magnetic circuit which has more than one path for flux is called a **parallel magnetic circuit**.

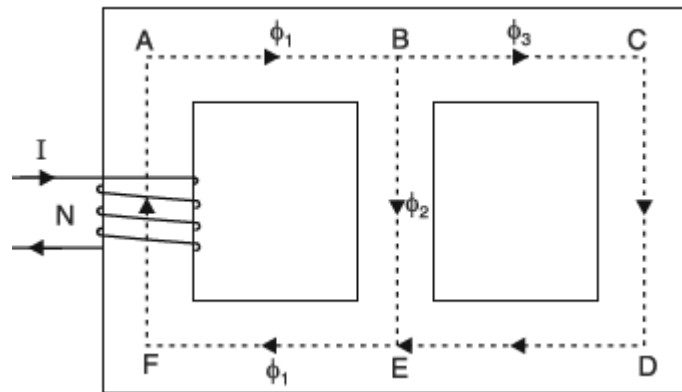


Fig.5.4.Parallel magnetic circuit

It can just be compared to a parallel electric circuit which has more than one path for electric current. The concept of parallel magnetic circuit is illustrated in Fig.5.4. Here a coil of N turns is wound on limb AF carries a current of I amperes. The flux Φ_1 set up by the coil divides at B into two paths, namely:

(i) flux Φ_2 passes along the path BE

(ii) flux Φ_3 follows the path $BCDE$

$$\text{Clearly, } \Phi_1 = \Phi_2 + \Phi_3$$

The magnetic paths BE and $BCDE$ are in parallel and form a parallel magnetic circuit. The AT required for this parallel circuit is equal to AT required for any one of the paths.

Let

$$\mathcal{R}_1 = \text{reluctance of path EFAB}$$

$$\mathcal{R}_2 = \text{reluctance of path BE}$$

$$\mathcal{R}_3 = \text{reluctance of path BCDE}$$

Total m.m.f. required = m.m.f. for path $EFAB$ + m.m.f. for path BE or path $BCDE$

$$NI = \Phi_1 \mathcal{R}_1 + \Phi_2 \mathcal{R}_2$$

or

$$NI = \Phi_1 \mathcal{R}_1 + \Phi_3 \mathcal{R}_3$$

The reluctances $\mathcal{R}_1$, $\mathcal{R}_2$ and $\mathcal{R}_3$ must be determined from a calculation of $l/\mu_0\mu_r$ for those

[Type here]

paths of the magnetic circuit in which Φ_1 , Φ_2 and Φ_3 exist respectively.

Exercise 2:

A cast steel magnetic structure made of a bar of section $2\text{ cm} \times 2\text{ cm}$ is shown in Fig. 5.5. Determine the current that the 500 turn magnetizing coil on the left limb should carry so that a flux of 2 mWb is produced in the right limb. Take $\mu_r = 600$ and neglect leakage.

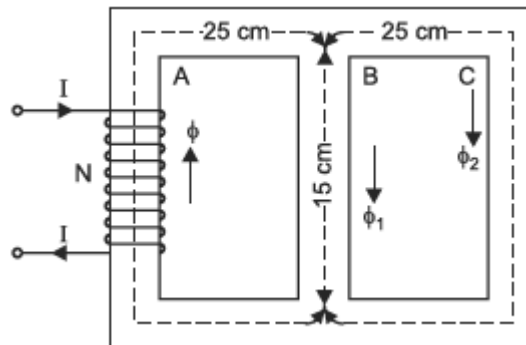


Fig.5.5 Exercise 2

Solution:

Step-1: Note the data given

Area of cross-section= $a=2\text{ cm} \times 2\text{ cm}=4\text{ cm}^2$

Number of turns= $N=500$

Flux in right limb= $\Phi=2\text{ mWb}$

$\mu_r = 600$

Step-2: To calculate total flux required

MMF required for the middle section 15 cm = MMF required for the right section 25 cm

$$\Phi_1 \left[\frac{\ell_1}{a_1 \mu_0 \mu_{r1}} \right] = \Phi_2 \left[\frac{\ell_2}{a_2 \mu_0 \mu_{r2}} \right]$$

$$\Phi_1 \ell_1 = \Phi_2 \ell_2$$

$$\Phi_1 = \Phi_2 \frac{\ell_2}{\ell_1} = 2 \times 10^{-3} \times \frac{25 \times 10^{-2}}{15 \times 10^{-2}} = 3.33 \text{ mWb}$$

$$\text{Total flux} = \Phi = \Phi_1 + \Phi_2 = 3.33 \times 10^{-3} + 2 \times 10^{-3} = 5.33 \times 10^{-3} \text{ Wb}$$

Step-3: To calculate total MMF required for left section and middle section

$$\text{MMF required for the left section } 25\text{ cm} = \Phi \left[\frac{l}{a \mu_0 \mu_{r1}} \right]$$

[Type here]

$$= 5.33 \times 10^{-3} \left[\frac{25 \times 10^{-2}}{4 \times 10^{-4} \times (4\pi \times 10^{-7}) \times 600} \right] = 4420 \text{ AT}$$

$$\text{MMF required for the middle section } 15\text{cm} = \phi_1 \left[\frac{\ell_1}{a_1 \mu_0 \mu_{r1}} \right]$$

$$= 3.33 \times 10^{-3} \left[\frac{15 \times 10^{-2}}{4 \times 10^{-4} \times (4\pi \times 10^{-7}) \times 600} \right] = 1657 \text{ AT}$$

Step-4: Total MMF required

Total MMF required

= MMF required for the left section 25cm+ MMF required for the middle section 15cm

or

= MMF required for the left section 25cm+ MMF required for the right section 25cm

Total MMF required

= MMF required for the left section 25cm+ MMF required for the middle section 15cm

=4420+1657

Total MMF required =6077

Step-5: To determine magnetizing current required

NI=6077

500I=6057

I=6057/500

I=12.11A

Hysteresis & B-H Curve Concepts:

Hysteresis, in the context of magnetism, describes the phenomenon where a material's magnetization lags behind the applied magnetic field. The B-H curve (Fig.5.6), or hysteresis loop, is a graphical representation of this relationship, plotting magnetic flux density (B) against magnetic field strength (H). This curve provides valuable insights into a material's magnetic properties, including its ability to retain magnetization and its resistance to demagnetization.

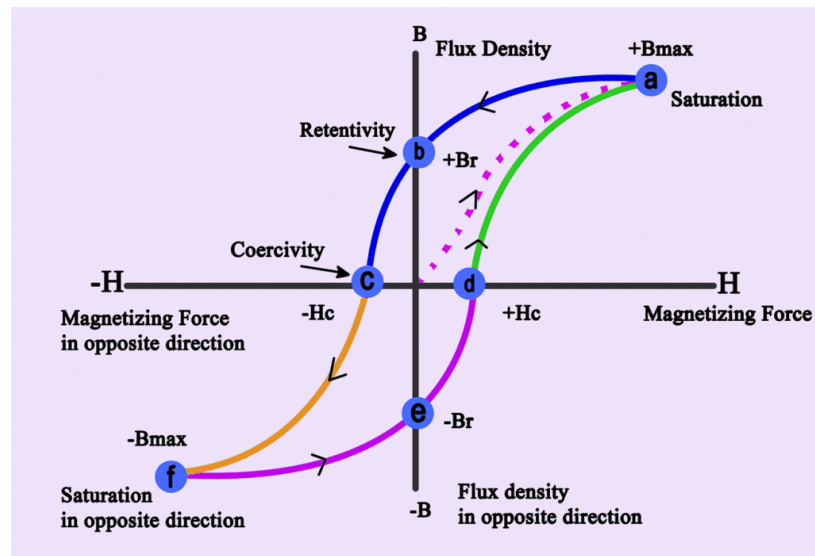


Fig.5.6.BH-Curve

When a ferromagnetic material is subjected to a changing magnetic field, its magnetization doesn't instantly respond to the field's strength. Instead, it lags behind, meaning the material retains some degree of magnetization even when the field is removed.

The B-H curve is a visual representation of this lagging effect. As the magnetic field (H) is applied and cycled, the resulting magnetic flux density (B) traces a loop. This loop is characteristic of the material and its magnetic behaviour and is known as **Hysteresis Loop**:

Faraday's laws of Electromagnetic Induction (EMI):

There are two laws of EMI:

First law:

It states that "whenever flux linking to the conductor changes, an EMF will be induced in the conductor. This EMF is called **statically induced EMF**"

Example: Alternator, Transformer and choke coil

Or

"When a conductor cuts magnetic field, an EMF will be induced in the conductor. This EMF is called **dynamically induced EMF**"

Example: DC generator

Second law:

It states that "The magnitude of EMF induced (e) in the coil is the rate of change of flux linkage ($N\Phi$)"

Mathematically

$$e = - \frac{d(N\Phi)}{dt} \text{ Volts}$$

[Type here]

Negative sign is due to Lenz's law which states that the induced current will flow in a direction that opposes the change in magnetic flux that produced it.

5.5 Statically induced emf:

Self-induced EMF:

Self-induced emf, also known as back emf, is the electromotive force (voltage) induced in a coil due to the changing magnetic flux produced by the current flowing through the coil itself.

This phenomenon occurs when the current in a circuit changes, causing a corresponding change in the magnetic field, which then induces a voltage opposing the change in current as in Fig.5.7

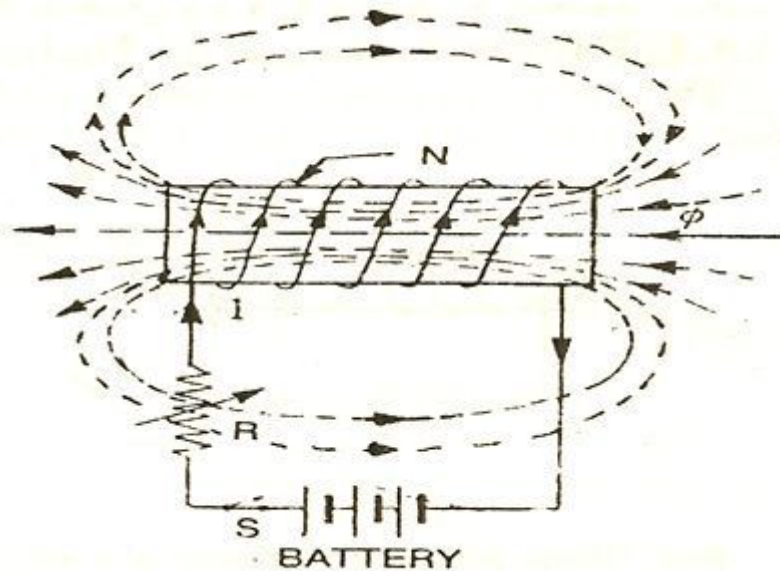


Fig.5.7: Concept of self-induced EMF

5.5.2 Mutually induced EMF:

Mutually induced EMF refers to the phenomenon where a changing current in one coil induces an electromotive force (EMF) in a nearby coil. This occurs due to the magnetic field produced by the first coil linking with the second coil, and a change in the magnetic flux induces an EMF in the second coil. The magnitude of the induced EMF is proportional to the rate of change of current in the first coil and the mutual inductance between the two coils as in Fig.5.8

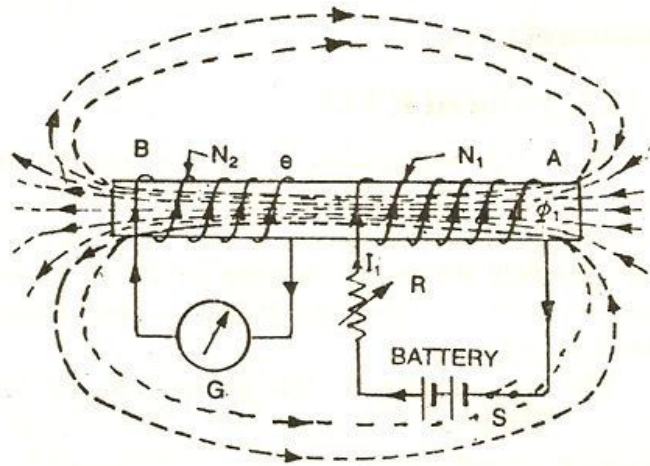


Fig.5.8: Concept of mutually-induced EMF

Example: Transformer

5.2. Transformer:

Definition:

It is a static electrical machine which transfers power from primary winding to secondary winding without change in frequency. In a transformer, voltage can be stepped up or stepped down. The basic principle of the transformer is mutual induction

Construction of transformer:

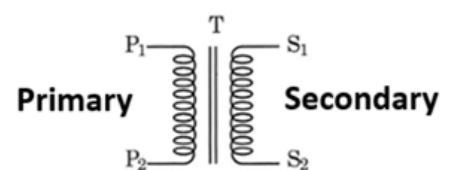
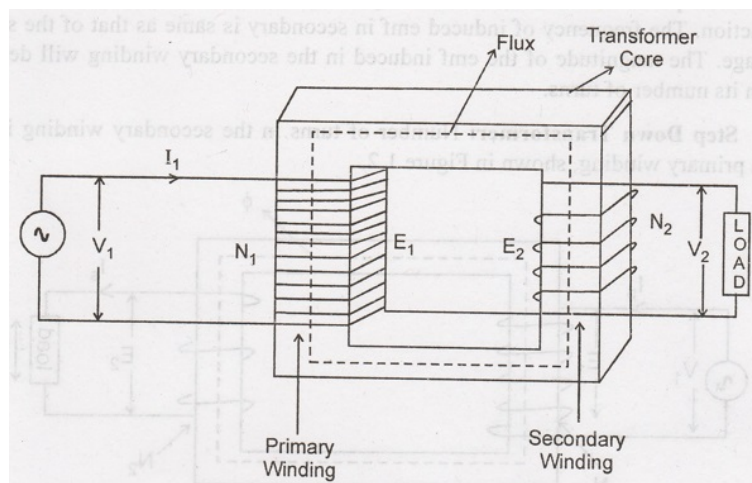


Fig.5.9: Transformer

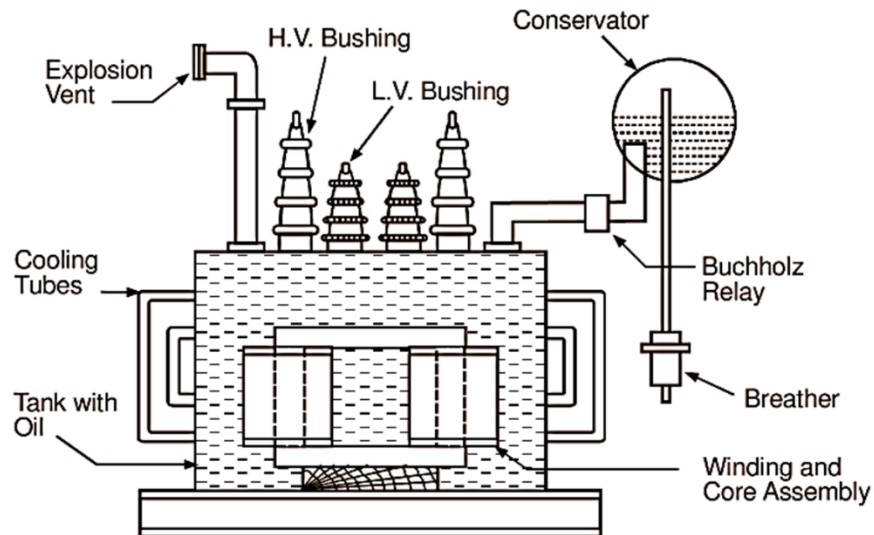


Fig.5.10: Practical Transformer

There are three major components of a Transformer:

- Iron Core
- Primary Winding
- Secondary Winding

Core:

The core of the transformer is rectangular in shape and laminated. It provides a low-resistance path for the magnetic flux and minimizes energy loss due to eddy currents. Laminated sheets are stacked to minimize air gaps and reduce reluctance. They are often E and I shaped, or in a stepped construction to maximize space utilization.

Windings:

There are two windings on the transformer i.e. Primary Winding and Secondary Winding. The Primary Winding is connected to the input terminal and is responsible for generating a self-induced EMF. The Secondary Winding is connected to the output load. These windings are placed on the core and are electrically insulated from each other and the core for proper functioning and reduction in losses.

These coils have different numbers of turns compared to each other. The Primary Winding of the Transformer has **N₁** turns. Similarly, the Secondary Winding of the Transformer has **N₂** turns. The relationship between the Primary side voltage and number of turns to

that of voltage is given by $\frac{V_1}{V_2} = \frac{N_1}{N_2}$

If $N_1 > N_2$ - Step down transformer- Steps down the voltage

If $N_1 < N_2$ - Step up transformer- Steps up the voltage

[Type here]

Transformer is usually placed inside a tank filled with oil. Bushings are provided for the insulated connections for the windings.

Exercise 3:

A sinusoidal flux 0.2 Wb (max.) links with 55 turns of a transformer secondary coil. Calculate the r.m.s. value to the induced EMF in the secondary coil. The supply frequency is 50 Hz.

Solution:

Given: $\Phi_m = 0.2 \text{ Wb}$, $N_2 = 55$, $E_2 = ?$, $f = 50 \text{ Hz}$

$$E_2 = 4.44 \Phi_m f N_1 = 4.44 * 0.2 * 50 * 55 = 2442 \text{ V}$$

Dynamically induced EMF:

Dynamically induced EMF, also known as motional EMF, is the electromotive force (EMF) generated in a conductor when it moves through a magnetic field, causing a change in magnetic flux linkage. This phenomenon is a core principle behind electrical generators and motors as in Fig.5.11

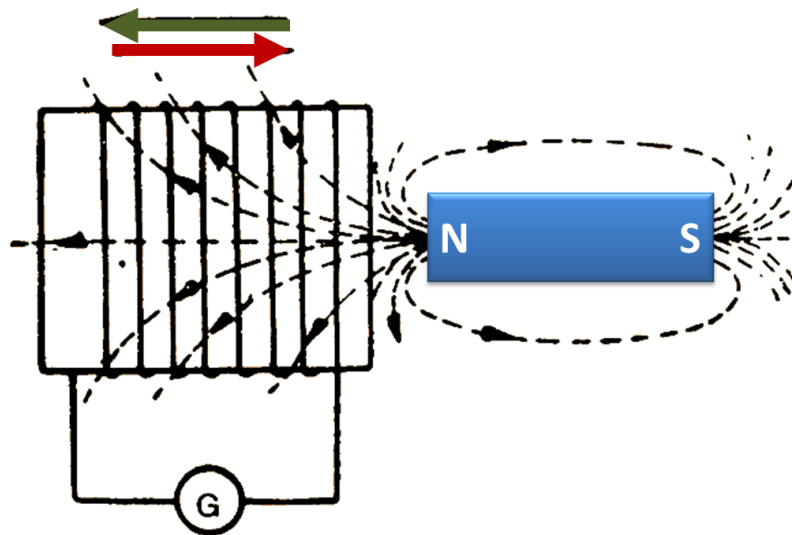


Fig.5.11: Concept of dynamically induced EMF

Example: DC Generator and DC Motor

Fleming's right-hand rule:

Fleming's right-hand rule helps determine the direction of induced current in a conductor moving within a magnetic field. It's a tool primarily used in generators to understand how electricity is produced.

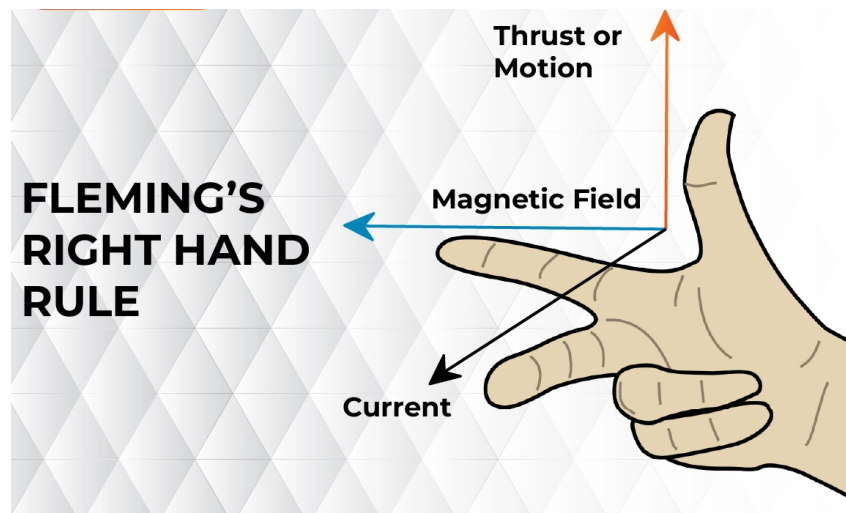


Fig.5.12. Fleming's Right Hand Rule

When three fingers i.e thumb, forefinger, and middle finger of right hand are held mutually perpendicular (at right angles) to each other, then the **thumb points** it in the direction of the conductor's motion (relative to the magnetic field), the **forefinger points** in the direction of the magnetic field, the **middle finger** indicates the direction of the induced current in the conductor.

Fleming's left-hand rule:

Fleming's left-hand rule is a mnemonic used to understand the relationship between the magnetic field, current, and force in a motor. It's a tool primarily used in motors to understand how mechanical force is produced.

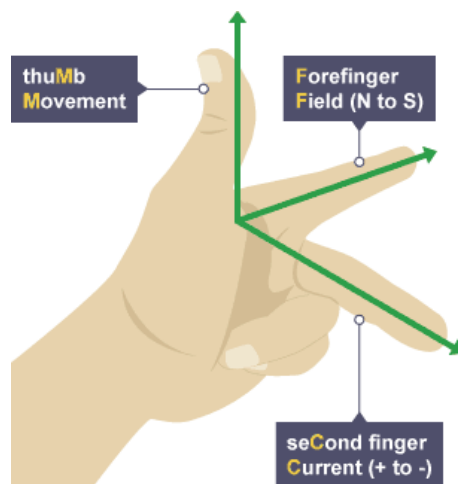


Fig.5.13. Fleming's Left Hand Rule

When three fingers i.e thumb, forefinger, and middle finger of left hand are held mutually perpendicular (at right angles) to each other, then the **index finger** points in the direction of the magnetic field, the **middle finger** points in the direction of the current, and the **thumb**

[Type here]

will point in the direction of the force (or motion) on the conductor

5.3 Electrical Machines

Concept of DC Machine:

A DC machine is **an electromechanical energy conversion device**. The DC machine can be classified into two types namely DC motors and DC generators.

DC Generator:

A DC generator works on the principle of electromagnetic induction. Specifically, it converts mechanical energy into direct current (DC) electrical energy by inducing a voltage in a conductor as it moves through a magnetic field. The induced voltage is dynamically induced EMF. This induced voltage then drives current through an external circuit.

DC Motor:

A DC motor converts electrical energy into mechanical energy through the interaction of magnetic fields and electric current. When a current-carrying conductor is placed in a magnetic field, it experiences a force, causing rotation. This principle, governed by Lorentz force and Fleming's left-hand rule, allows a DC motor to produce torque and motion.

Construction of DC Machine:

A DC machine, either a motor or generator, has two main parts: a stator and a rotor. The stator is the stationary part, and the rotor is the rotating part. Key components include the yoke (frame), poles, field windings, armature core, armature winding, commutator, and brushes.

Stator:

1.Yoke (Frame):

Provides mechanical support for other components and protects the machine from external disturbances. It also serves as a return path for the magnetic flux

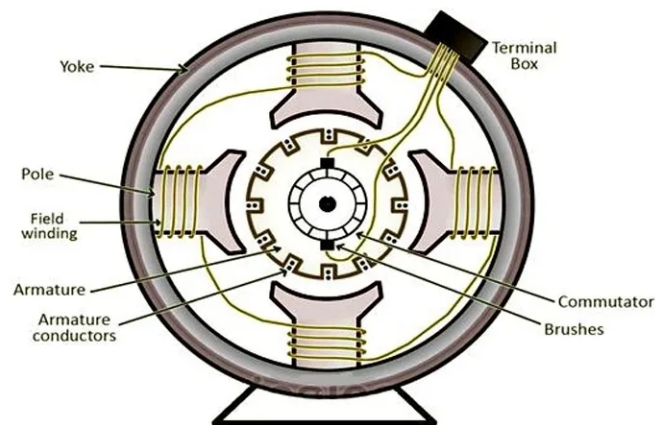


Fig.5.14: DC Machine

2.Poles and Pole Shoes:

Electromagnets that produce the magnetic field. They consist of a pole core and a pole shoe. Pole shoe help to spread the flux in the air gap.

3.Field Windings:

Coils of wire wound around the poles, creating a magnetic field when energized by a DC current.

4.Interpoles (Compensating Windings):

Smaller poles located between main poles, used to counteract armature reaction and improve commutation.

5.Armature Core: Cylindrical structure with slots to hold the armature winding. It is made of laminated silicon steel to reduce eddy current losses.

6.Armature Winding:

A system of conductors where the voltage is induced.

7.Commutator:

A segmented copper ring that reverses the current direction in the armature windings, ensuring unidirectional torque in motors or a steady DC output in generators.

8.Brushes:

Conductive elements (usually carbon or graphite) that maintain electrical contact with the commutator to transfer current.

Types of DC motors:

[Type here]

On the basis of the connections of armature and their field winding, DC motors can be classified as;

1. Separately excited DC motor:

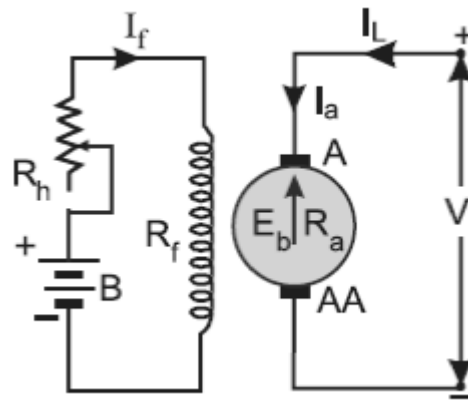


Fig.5.18. Separately excited DC motor -Circuit diagram

The conventional diagram of a separately excited DC motor is shown Fig. 5.18. Its voltage equation will be; $E_b = V - I_a R_a - 2V_b$ (Where) where

E_b = Back emf produced by the armature and is given by $E_b = \frac{\Phi Z N}{60} \times \frac{P}{A}$ where

Φ = armature flux in Wb, Z = number of conductors ,

A = number of parallel paths = 2 for wave and P for lap winding machine

V = Applied voltage

R_a = Armature resistance in Ω

V_b = voltage drop per brush in Volts

2. Self excited DC motors:

These motors can be further classified as;

(i) Shunt motor:

In Shunt wound DC Motor, the field windings are in parallel with armature as shown in Fig. 5.19.

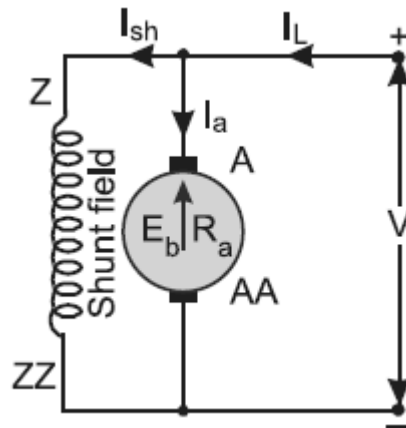


Fig.5.19 Self excited DC shunt motor -Circuit diagram

Equations governing the operation of a shunt motor are

$$I_{sh} = \frac{V}{R_{sh}}$$

$$I_a = I_L - I_{sh}$$

$$E_b = V - I_a R_a - 2V_b$$

Where,

I_{sh} = the field current

I_L = the line current

R_{sh} = the field resistance in Ω

R_a =Armature resistance in Ω

E_b = Back emf produced by the armature and is given by $E_b = \frac{\Phi Z N}{60} \times \frac{P}{A}$ where

Φ = armature flux in Wb, Z = number of conductors ,

A = number of parallel paths = 2 for wave and P for lap winding machine

V = Applied voltage

V_b = voltage drop per brush in Volts

(ii) **Series motor:** In series-wound DC Motor, the field windings are in series with the armature and is as shown in Fig. 5.20

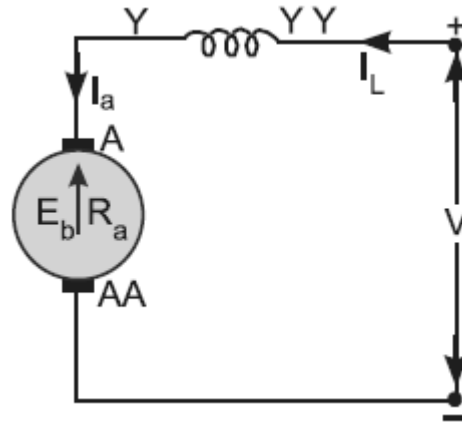


Fig.5.20. Self-excited DC series motor -Circuit diagram

Equations governing the operation of a series motor are

$$I_L = I_a = I_{se}$$

$$E_b = V - I(R_a + R_{se}) - 2V_b$$

Where,

I_{se} = the field current

I_L = the line current

R_{se} = the field resistance in Ω

R_a =Armature resistance in Ω

E_b = Back emf produced by the armature and is given by $E_b = \frac{\Phi Z N}{60} \times \frac{P}{A}$ where

Φ = armature flux in Wb, Z = number of conductors ,

A = number of parallel paths = 2 for wave and P for lap winding machine

V = Applied voltage

V_b = voltage drop per brush in Volts

Exercise 4: problem on shunt motor

A 220V DC shunt motor has an armature resistance of 0.5Ω and shunt field resistance of 110Ω . If the full load line current is 20 A, find the induced back EMF, input power and the Power developed

Solution:

Step-1: Note down the given data

Given: $V=220V$, $R_a=0.5\Omega$, $R_{sh}=110\Omega$, $I_L=20A$, E_b , P_{in} , P_d

Step-2: Determine shunt field current and armature current

[Type here]

$$\text{Shunt field current} = I_{sh} = \frac{V}{R_{sh}} = \frac{220}{110} = 2A$$

$$\text{Armature current} = I_a = I_L - I_{sh} = 20 - 2 = 18A$$

Step-3: Determine back EMF

$$\text{Back EMF} = E_b = V - I_a R_a = 220 - 18 \times 0.5 = 211V$$

Step-4: Determine input and output powers

$$\text{Power input} = P_{in} = V I_L = 220 \times 20 = 4400W$$

$$\text{Power developed} = P_d = E_b I_a = 211 \times 18 = 3798W$$

Exercise 5: problem on series motor

A 220V DC series motor has an armature resistance of 1Ω and series field resistance of 0.5Ω .

If the full load line current is 30 A, find the induced back EMF, input power and the Power developed

Solution:

Step-1: Note down the given data

Given: $V=220V$, $R_a=1\Omega$, $R_{se}=0.5\Omega$, $I_a=I_{se}=I_L=30A$, E_b , P_{in} , P_d

Step-2: Determine back EMF

$$\text{Back EMF} = E_b = V - I_a R_a - I_{se} R_{se} = 220 - 30 \times 1 - 30 \times 0.5 = 175V$$

Step-4: Determine input and output powers

$$\text{Power input} = P_{in} = V I_L = 220 \times 30 = 6600W$$

$$\text{Power developed} = P_d = E_b I_a = 175 \times 30 = 5250W$$

Stepper Motor

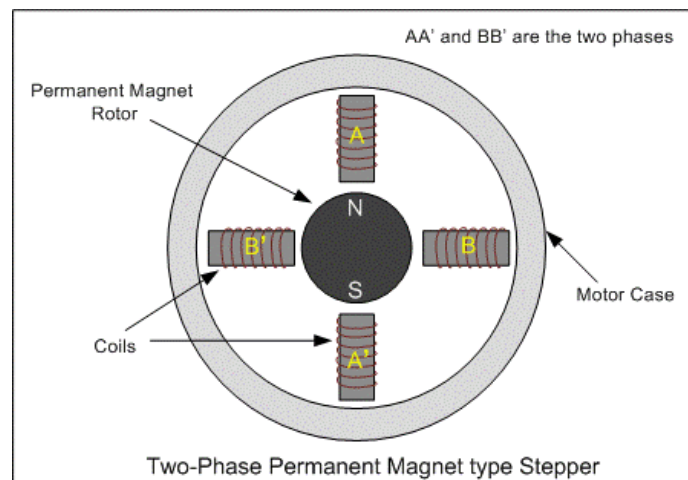


Fig.5.21: Stepper Motor

[Type here]

A **stepper motor** works by converting **electrical pulses into precise mechanical movement**. Instead of rotating smoothly like a DC motor, it moves in **discrete steps**, which is why it's great for precise position control.

Working principle:

A stepper motor has **stator coils (electromagnets)** and a **rotor** (permanent magnet or soft iron). When the stator coils are energized in a **specific sequence**, they create a **rotating magnetic field**. The rotor aligns itself with this magnetic field. Each change in the energized coil makes the rotor move **one fixed step**.

One electrical pulse = one mechanical step. The **step angle** is the angle the motor moves per pulse. Example: 1.8° step angle $\rightarrow$ **200 steps per full revolution** Smaller step angle = **higher precision**

Types of stepper motors:

i) **Permanent Magnet (PM) Stepper:** Rotor is a permanent magnet Simple and low cost

ii) **Variable Reluctance (VR) Stepper:** Rotor is soft iron Moves to position of least magnetic resistance

iii) **Hybrid Stepper:** Combines PM and VR features Most common, high accuracy

Applications of stepper motors

1) Industrial applications:

- **CNC machines** – precise control of tool position
- **3D printers** – accurate movement of print head and bed
- **Robotics** – joint and arm positioning
- **Packaging machines** – indexing and timing operations
- **Textile machines** – controlled fabric movement

2) Computer and Office applications

- **Printers** (inkjet, laser) – paper feed and print head motion
- **Scanners** – precise scanning movement
- **Disk drives** – head positioning (older HDDs, floppy drives)

3) Medical applications:

[Type here]

- **Infusion pumps** – accurate fluid delivery
- **Syringe pumps**
- **Medical imaging equipment** – controlled motion mechanisms

4)Automated lab instruments Consumer applications:

- **Camera lens focusing and zoom mechanisms**
- **DVD/CD drives**
- Smart locks

5)Automotive applications

- Speedometers and tachometers
- Fuel injection systems
- HVAC control actuators
- Headlight leveling systems

6)Aerospace and defense applications

- Satellite positioning systems
- Antenna positioning
- Missile guidance mechanisms

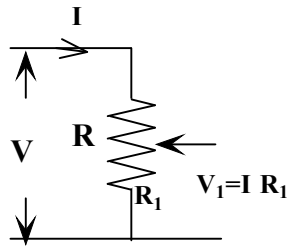
7)Home and automation applications:

- Automatic curtains and blinds
- Vending machines
- ATM machines
- Washing machines

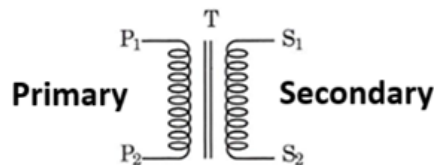
5.4 Coupled Circuits

Two circuits are said to be 'coupled' when energy transfer takes place from one circuit to the other when one of the circuits is energized. There are many types of Coupled Circuits

i)conductive coupling. Example: potential divider

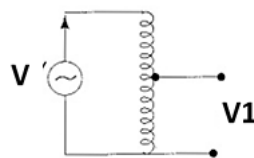


ii) inductive or magnetic coupling Example: two winding transformers



iii) Both conductive and
Example: auto transformer

inductive coupling



Magnetically Coupled Circuits:

When two coils are placed very close to each other, a changing flux in one coil induces voltage in second coil. Two coils are said to be mutually coupled coils.

The coefficient of coupling:

(denoted by 'k') quantifies the degree of magnetic coupling between two coils. It represents the fraction of magnetic flux produced by one coil that links with the other coil. A coefficient of 1 (or 100%) indicates perfect coupling where all flux links, while 0 means no flux is linked. Consider the two coils having self-inductances L_1 and L_2 and mutual inductance M as shown in Fig 5.19.

Let

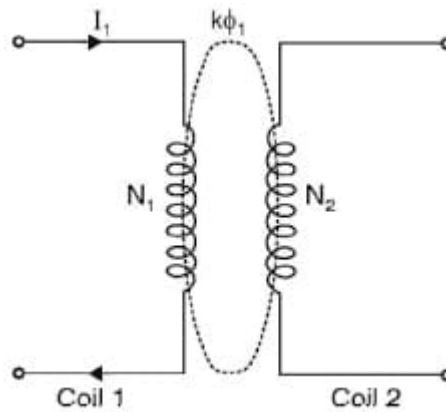
Φ_1 be the total flux in the coil 1 in Wb

Φ_2 be the total flux in the coil 2 in Wb

Φ_{12}, Φ_{21} be the linking flux between coil 1 and coil 2 in Wb

M_{12}, M_{21} are the mutual inductance between coil 1 and coil 2 = M

[Type here]

**Fig.5.19: Coupling coils****Coefficient of coupling (k)**

linking flux to total flux

is defined as the ratio of

$$\text{Coefficient of coupling} = k = \frac{\Phi_{12}}{\Phi_1} = \frac{\Phi_{21}}{\Phi_2}$$

Expression for coefficient of coupling (k):

$$\text{Coefficient of coupling} = k = \frac{\Phi_{12}}{\Phi_1} = \frac{\Phi_{21}}{\Phi_2}$$

$$M_{12} = M = \frac{N_2 \Phi_{12}}{i_1} = \frac{N_2 k \Phi_1}{i_1} \text{ for coil-2}$$

$$M_{21} = M = \frac{N_1 \Phi_{21}}{i_2} = \frac{N_1 k \Phi_2}{i_2} \text{ for coil-1}$$

$$L_1 = \frac{N_1 \Phi_1}{i_1} \text{ for coil-1}$$

$$L_2 = \frac{N_2 \Phi_2}{i_2} \text{ for coil-2}$$

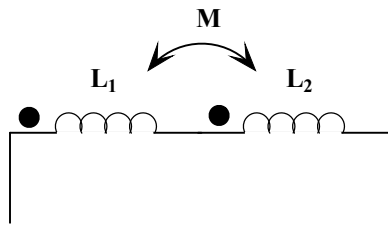
$$M_{12} * M_{21} = M * M = \frac{N_2 k \Phi_1}{i_1} * \frac{N_1 k \Phi_2}{i_2} = k^2 \frac{N_1 \Phi_1}{i_1} * \frac{N_2 \Phi_2}{i_2} = k^2 L_1 L_2$$

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

5.5 Dot convention:

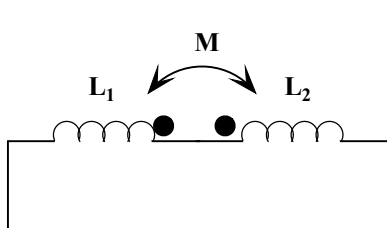
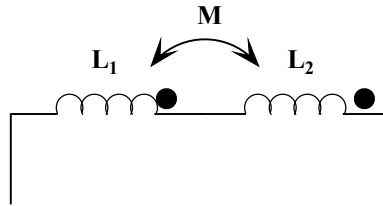
The dot convention in coupled circuits indicates the polarity of induced voltage in magnetically coupled coils. It's a visual aid, using dots on the coils, to determine whether the mutual flux adds or subtracts, thus influencing the direction of induced currents. Consider the four cases shown in Fig 5.20

[Type here]



Total inductance of the circuit

$$L = L_1 + M + L_2 + M = L_1 + L_2 + 2M$$



Total inductance of the circuit

$$L = L_1 - M + L_2 - M = L_1 + L_2 - 2M$$

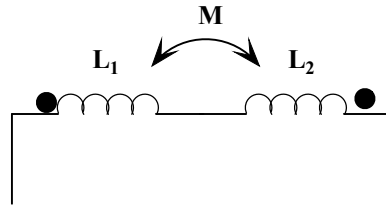
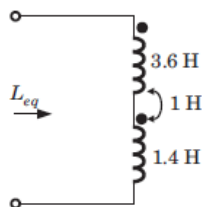


Fig.5.20: Dot conventions

Exercise 6:

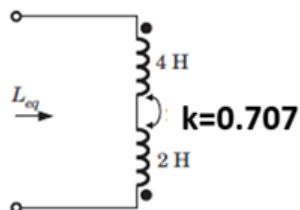
Find the equivalent inductance of the circuit shown below



$$L_{eq} = 3.6 + 1 + 1.4 + 1 = 7 \text{ H}$$

Exercise 7:

Find the equivalent inductance of the circuit shown below



$$M = k\sqrt{L_1 L_2}$$

[Type here]

$$M = 0.707\sqrt{4 \times 2} = 2 \text{ H}$$

$$L_{eq} = 4 - 2 + 2 - 2 + 1 = 2 \text{ H}$$

Exercise 8:

Calculate the coefficient of coupling having self-inductances of 50 mH and 60 mH. The mutual inductance between them is 30 mH

$$L_1 = 50 \text{ mH}, L_2 = 60 \text{ mH}, M = 30 \text{ mH}$$

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{30 \times 10^{-3}}{\sqrt{50 \times 10^{-3} \times 60 \times 10^{-3}}} = 0.5477$$

Mesh analysis of coupled circuits:**Exercise 9:**

Two coupled coils with $L_1 = 0.5 \text{ H}$ and $L_2 = 4 \text{ H}$ have a coefficient of coupling 0.8. Find the maximum value of inductance EMF in the coil-2 if a current of $i_1 = 20 \sin 314t$ passed in the coil-1

Step1: Determine mutual inductance

$$M = k\sqrt{L_1 L_2} = 0.8\sqrt{0.5 \times 4} = 1.13$$

Step2: Determine instantaneous mutually induced EMF in coil-2

$$e_{M2} = M \frac{di_1}{dt} = 1.13 \frac{d(20 \sin 314t)}{dt} = 1.13(314 \times 20 \cos 314t) = 7096.2 \cos 314t$$

Step3: Obtain the maximum value of inductance EMF

The maximum value of inductance EMF = 7096.2 V

Exercise 10:

Find the voltage across the 10Ω resistor for the network shown in Fig 5.21. Take $X_M = j3 \Omega$

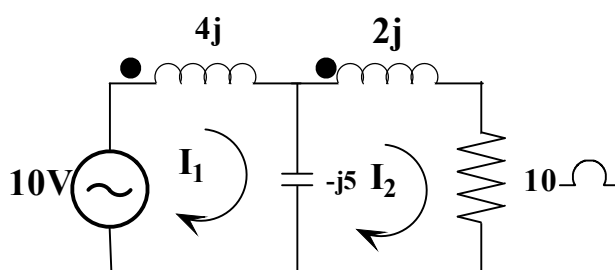


Fig 5.21 Exercise 10

Applying KVL to mesh 1,

$$-4j \times I_1 + 5j \times (I_1 - I_2) + 10 - 3j \times I_2 = 0$$

[Type here]

$$jxI_1 - 8jxI_2 = -10$$

$$-jxI_1 + 8jxI_2 = 10$$

Applying KVL to mesh 2,

$$-2jxI_2 - 10xI_2 + 5jx(I_2 - I_1) - 3jxI_1 = 0$$

$$-8jxI_1 - (10 - 3j)xI_2 = 0$$

$$8jxI_1 + (10 - 3j)xI_2 = 0$$

Applying Cramm's rule,

$$I_2 = \frac{\begin{vmatrix} -j1 & 10 \\ 8j & 0 \end{vmatrix}}{\begin{vmatrix} -j1 & 8j \\ 8j & 10 - 3j \end{vmatrix}} = \frac{-80j}{61 - 10j} = 0.203 - 1.28j = 1.29\angle -81^\circ \text{ A}$$

Voltage across 10Ω resistor is $10 \times 1.29\angle -81^\circ = 12.9\angle -81^\circ \text{ V}$

Exercise 11:

Calculate the voltage V_2 in the circuit of Figure 5.22. A

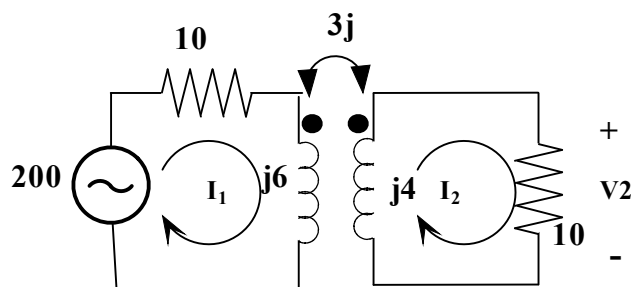


Fig 5.22 Exercise 11

Applying KVL to mesh 1,

$$-10xI_1 - j6xI_1 + 3jxI_2 + 200 = 0$$

$$-(10+j6)jxI_1 + 3jxI_2 = -200$$

$$(10+j6)jxI_1 - 3jxI_2 = 200$$

Applying KVL to mesh 2,

[Type here]

$$-10xI_2 - 4jxI_2 + 3jxI_1 = 0$$

$$3jxI_1 - (10 + 4j)xI_2 = 0$$

Applying Cramm's rule,

$$I_2 = \frac{\begin{vmatrix} (10 + 6j) & 200 \\ 3j & 0 \end{vmatrix}}{\begin{vmatrix} (10 + 6j) & -3j \\ 3j & -(10 + 4j) \end{vmatrix}} = \frac{-600j}{-85 - 100j} = 3.48 + 2.96j = 4.57 \angle 40.4^\circ \text{ A}$$

Voltage across 10Ω resistor is $V_2 = 10 \times 4.57 \angle 40.4^\circ = 45.7 \angle 40.4^\circ \text{ V}$

5.6 Protective Devices

5.6.1 Fuse:

A fuse is an electrical safety device that protects a circuit by melting and breaking the circuit when excess current flows

Function:

- It contains a **thin metal wire or strip** (fuse element).
- When current exceeds the safe limit, the element **heats up and melts**.
- This **breaks the circuit**, stopping current flow.
- Result: equipment is protected from overheating, fire, or damage.

Working Principle:

- Based on the **heating effect of electric current**.
- Excess current $\rightarrow$ more heat $\rightarrow$ fuse element melts $\rightarrow$ circuit opens.

Main Parts:

- Fuse element (metal wire/strip)
- Terminals
- Insulating body (glass, ceramic, or porcelain)

Types of Fuses:

- **Rewirable (Kit-Kat) Fuse** – used in homes

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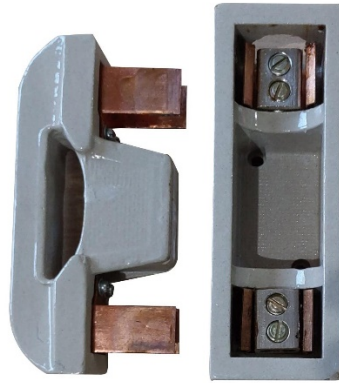


Fig.5.23: Kit-Kat

- **Cartridge Fuse** – glass or ceramic, common in electronics



Fig.5.24: Cartridge Fuse

- **HRC Fuse (High Rupturing Capacity)** – industrial use



Fig.5.25: HRC Fuse

- **Automotive Fuse** – vehicles



Fig.5.26: Automotive Fuse

- **Thermal Fuse** – temperature-based protection



Fig.5.27: Thermal Fuse

Applications:

- Household wiring
- Electronic devices (TVs, chargers, power supplies)
- Automobiles
- Industrial control panels

[Type here]

- Power distribution systems

Advantages:

- Simple and inexpensive
- Reliable protection
- No maintenance required

Disadvantages

- Must be **replaced after blowing**
- Not reusable
- Slower than circuit breakers in some cases

5.62 Miniature Circuit Breaker (MCB):

A Miniature Circuit Breaker **is an** automatic electrical safety device that protects a circuit from:

- **Overload** (too many appliances drawing current)
- **Short circuit** (live and neutral touch)

When a fault happens, the MCB **trips** and cuts off power, preventing:

- Wire overheating
- Fire hazards
- Damage to appliances

Working Principle:

- **Overload:** A bimetallic strip heats up, bends, and trips the breaker.
- **Short circuit:** An electromagnetic mechanism trips it **instantly**.

Types of MCBs:

Type B – Trips at 3–5 times rated current

- Homes, lighting, sockets



Fig.5.28: Type-B MCB

Type C – Trips at 5–10× times rated current

- Motors, air conditioners



Fig.5.29: Type-C MCB

Type D – Trips at 10–20× times rated current

- Heavy industrial equipment



Fig.5.29: Type-D MCB

MCB ratings:

- **Current rating:** 6A, 10A, 16A, 20A, 32A, etc.
- **Voltage rating:** Usually 230V (single phase) or 415V (three phase)

Breaking capacity: 6kA or 10kA (fault current it can safely break)

MCB vs Fuse:

- MCB Reusable (just reset it)
- MCB Faster and more reliable
- MCB Safer and cleaner than fuses

5.7 Earthing:

5.71 Necessity:

Earthing is necessary to protect the human body from electric shock and to maintain electrical balance with the Earth

5.72 Types of earthing:

- 1) Pipe earthing
- 2) Plate earthing

1) Pipe earthing:

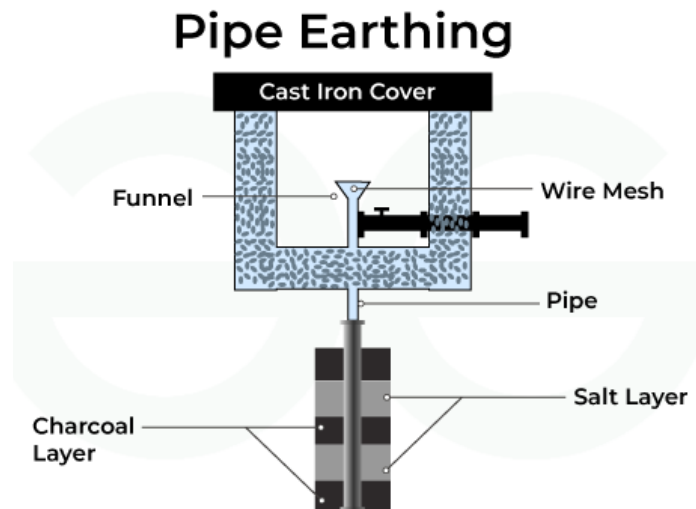


Fig.5.30: Pipe earthing

Pipe earthing is a highly efficient, cost-effective, and common method of electrical grounding that uses a vertically buried, perforated Galvanized Iron (GI) pipe, typically 38-75 mm in diameter and 2.5-3 meters long, to direct fault current into the ground. It is ideal for low-to-medium voltage applications and requires, for low-resistance, alternating layers of charcoal and salt surrounding the pipe, along with regular water pouring for maintenance

2)Plate earthing:

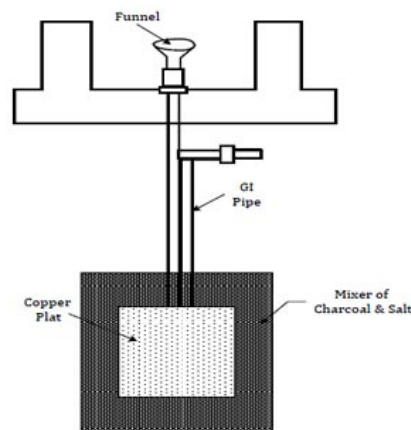


Fig.5.31:Plate earthing

Plate earthing is a, reliable,, and common method of electrical grounding that involves burying a, conductive, plate, (typically, copper, or, galvanized, iron), vertically, at, least, 3, meters, (about, 8-10, feet), below, ground, level, to, safely, dissipate, fault, currents, into, the, earth,. This, technique, utilizes, a, 600mm x 600mm, plate, often, surrounded, by, alternating, layers, of, salt, and, charcoal, along with regular water pouring for maintenance

[Type here]

to, maintain, low, soil, resistivity

5.73 Ground Concept:

Ground in electricity is a fundamental concept representing a zero-voltage reference point and a safe, low-resistance return path for current to the physical earth. It protects against electric shock and equipment damage by directing excess, stray, or fault current away from users and into the ground

5.74 Neutral Concept:

In electrical systems, the **neutral** is a conductor that provides the return path for current back to the power source, completing the circuit. It is typically connected to ground at the main service panel, maintaining a reference point near zero voltage, which distinguishes it from the live (hot) wire
